北京郵電大學
Beijing University of Posts and Telecommunications

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1.

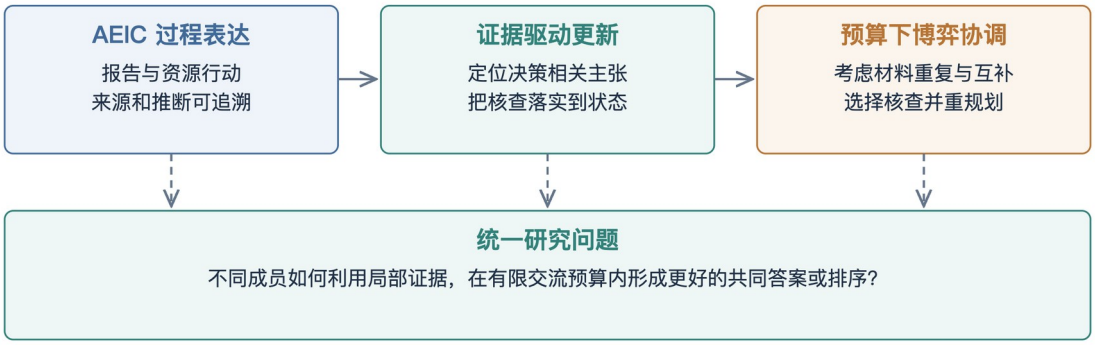
[1–4]

2.

AEIC

1

研究递进与当前问题



评价链条：信息有用 → 选择有效 → 更新准确 → 最终决策改善

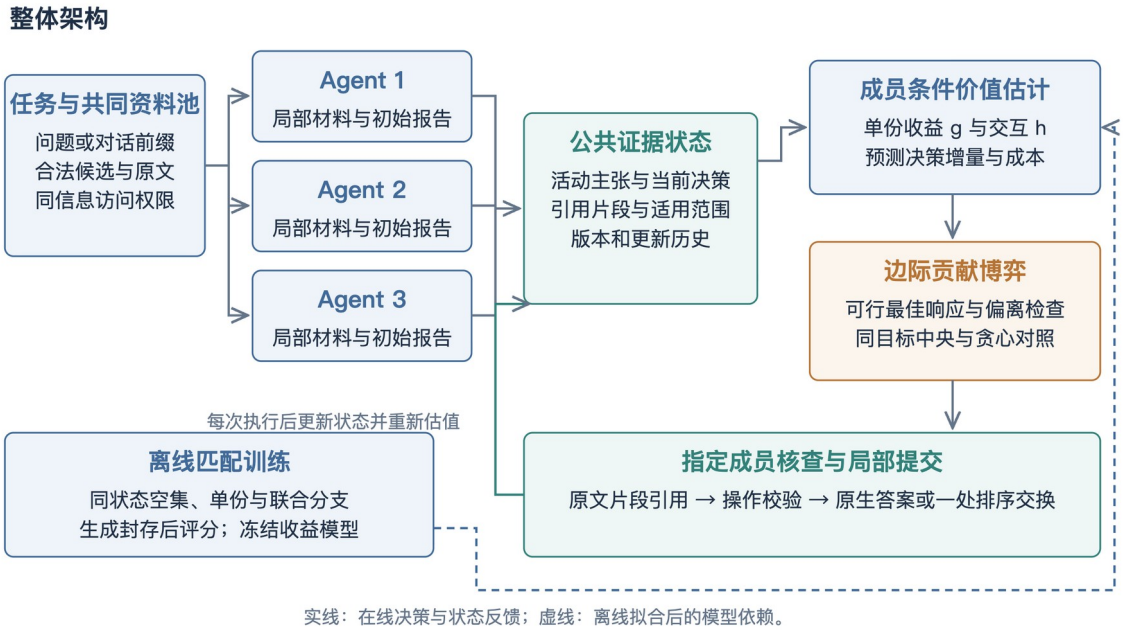
三个阶段构成一条研究主线；排序器进步与博弈协调收益分别归因。

1

— [3–6]

3.

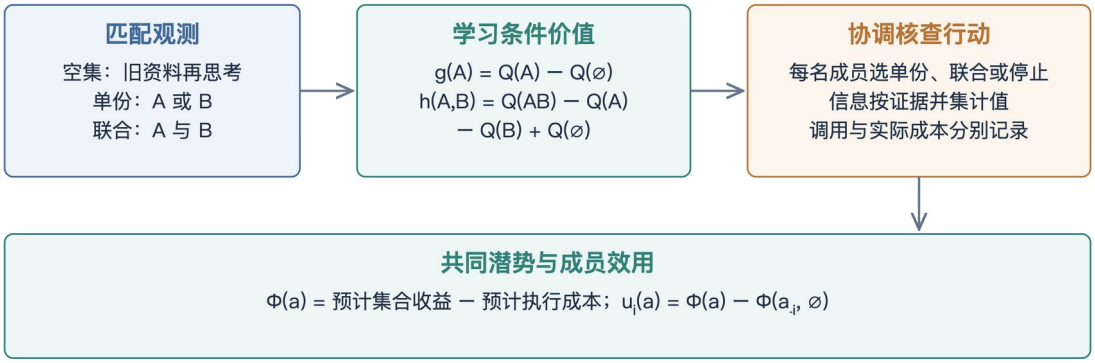
2


整体架构
任务与共同资料池
问题或对话前缀
合法候选与原文
同信息访问权限
Agent 1
局部材料与初始报告
Agent 2
局部材料与初始报告
Agent 3
局部材料与初始报告
公共证据状态
活动主张与当前决策
引用片段与适用范围
版本和更新历史
成员条件价值估计
单份收益 g 与交互 h
预测决策增量与成本
边际贡献博弈
可行最佳响应与偏离检查
同目标中央与贪心对照
指定成员核查与局部提交
原文片段引用 → 操作校验 → 原生答案或一处排序交换
每次执行后更新状态并重新估值
离线匹配训练
同状态空集、单份与联合分支
生成封存后评分；冻结收益模型
实线：在线决策与状态反馈；虚线：离线拟合后的模型依赖。

2

A B AB 3

集合价值与资源协调



固定状态：可行单边改进 → 无有利偏离 → 执行一个核查行动

状态变化：重新构造价值与博弈；有限预算控制整轮终止

均衡只针对预计目标；中央同目标最优与真实任务质量均需另外比较。

3

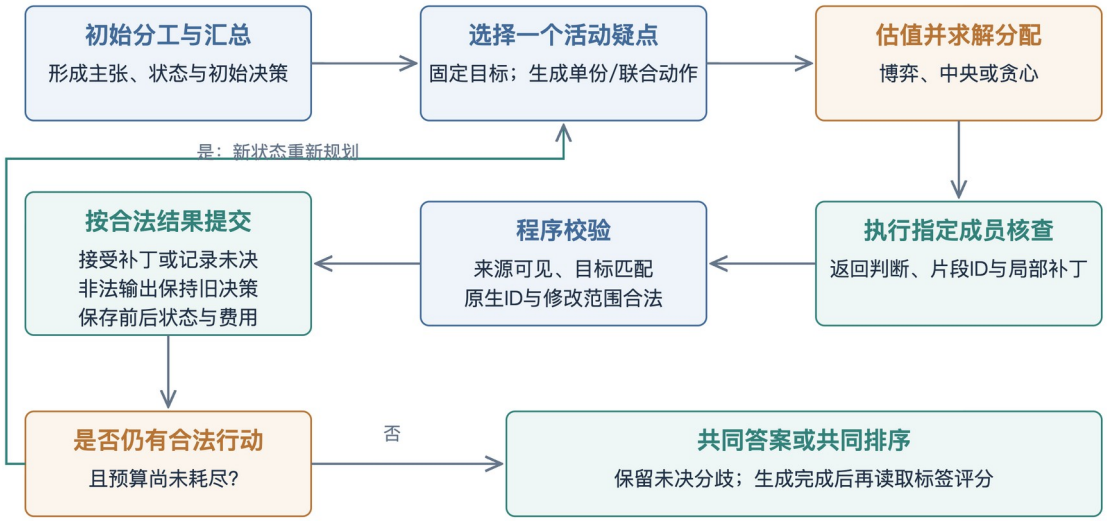
$$g_i(e \mid s) = Q(Y_e) - Q(Y\varnothing)$$

$$h_i(e,f \mid s) = Q(Y_{ef}) - Q(Y_e) - Q(Y_f) + Q(Y\varnothing)$$

$$u_i(a \mid s) = \Phi(a \mid s) - \Phi(a_{-i}, \varnothing \mid s)$$

4

在线核查与更新流程

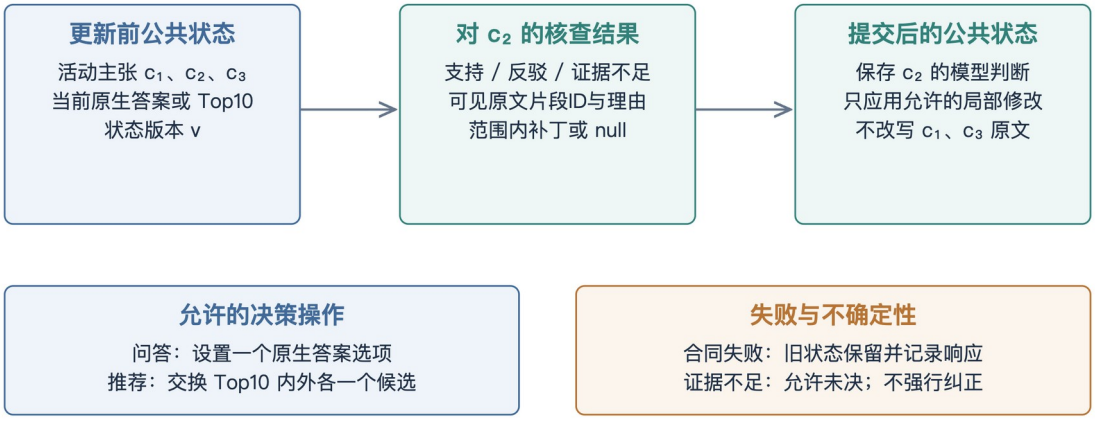


4

ID ID ID

5

受限更新与事务边界



结构与来源校验 ≠ 语义真实性认证

语义质量通过公开人工标签及任务结果独立评价；该图为操作范围示意。

5

Top10

1.

—

123 200 StrategyQA 200 ReDial 458 HotpotQA 180
64
47 17

400
HotpotQA 458

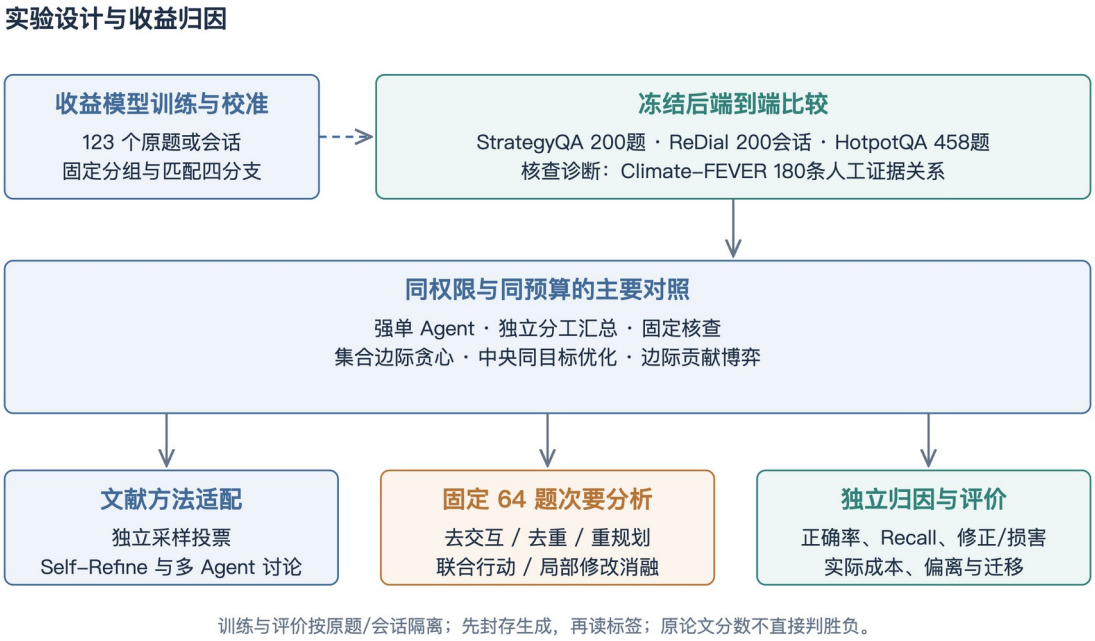
AEIC

ReDial 3010 806 730 Recall@10
24.252% 26.777% 2.525
64 83 21 22 20

8 256 85

6

Recall



6

StrategyQA HotpotQA 458 447
10 11 2 9

ReDial 200 Climate-FEVER 180 claim

400 116/200 58.0% 51/263
Recall@10 19.4% 0 [0,0]
Agent 122/200 61.0%
Recall@10 22.4% Sparse RRF60 KBRD 17.1% 16.7% 16.4%
200 2.5 0 3 21
28.1%

256

24 7 8

Agent

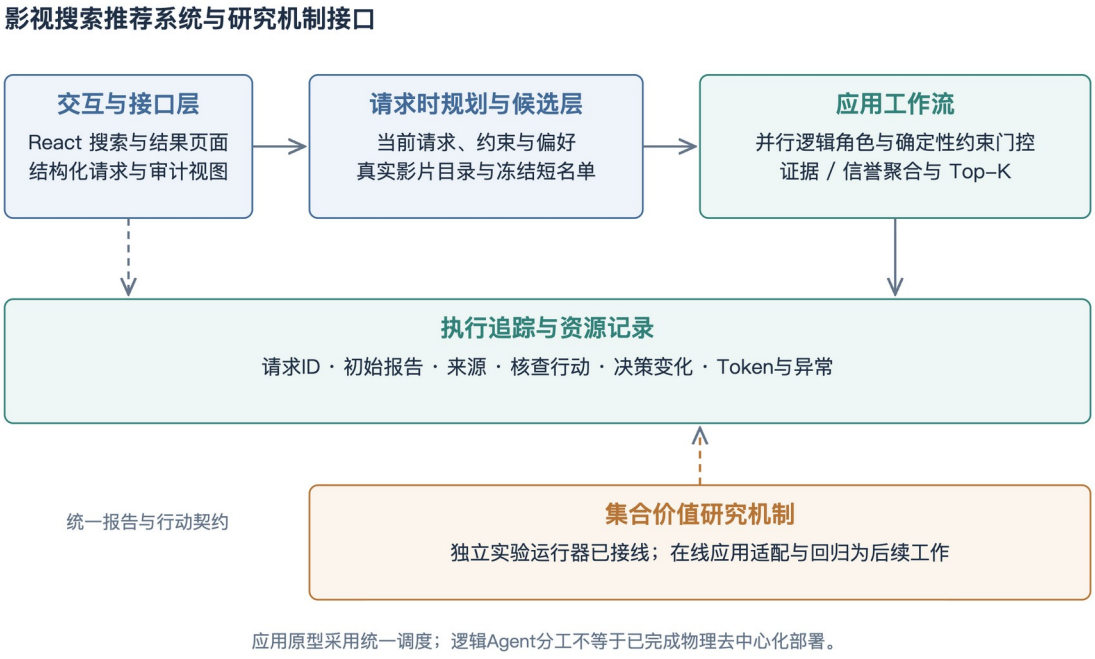
Agent
Agent
Self-Refine

Sparse KBRD RRF60 API Recall@10
17.1% 16.4% 16.7%

hash 64
400 458
QA ReDial 0.2 200
8.9 458 5.9
Token

React

HTTP 7



7

Top-K

Agent

ID

8



8

main

benchmark hash /

ID ReDial

ReDial Sparse/KBRD/RRF

17

ID

Token

1.

API 12

×

Agent Agent

CBBA

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